## Exercises

**Exercise 5.1 · Collinearity and orthogonality.** Consider the points $A$, $B$, $C$, $D$ with affixes $a = 1 + i$, $b = 3 + 2i$, $c = -1$ and $d = 2 - i$.

1. Compute $\dfrac{c-a}{b-a}$. What can be deduced?
2. Compute $\dfrac{d-a}{b-a}$. What can be deduced about the triangle $ABD$?

**Exercise 5.2 · Nature of triangles.** Determine the nature of the triangle $ABC$ in each case:

1. $a = 0$, $b = 2$, $c = 1 + i\sqrt{3}$;
2. $a = 1$, $b = 2 + i$, $c = i$.

**Exercise 5.3 · Sixth roots.** Determine the elements of $\mathbb{U}_6$ in exponential form, then in algebraic form. Check that their sum is zero.

**Exercise 5.4 · Cube roots of a complex number.** Solve in $\mathbb{C}$ the equation $z^3 = 8i$. Write $z = re^{i\theta}$ and give the solutions in algebraic form.

**Exercise 5.5 · Completing a square.** Let $A(1 + 2i)$ and $B(4 + 3i)$. Determine the affixes of the points $C$ and $D$ such that $ABCD$ is a direct square. Check that the diagonals have the same midpoint.

**Exercise 5.6 · Sets of points.** Let $A$ and $B$ be the points with affixes $2i$ and $-1$. To every point $M$ with affix $z \neq -1$, we associate $Z = \dfrac{z - 2i}{z + 1}$.

1. Interpret geometrically the modulus and an argument of $Z$ when $z \neq 2i$.
2. Determine the set of points $M$ such that $Z$ is real.
3. Determine the set of points $M$ such that $Z$ is purely imaginary and nonzero.
4. Determine the set of points $M$ such that $|Z| = 1$.

**Exercise 5.7 · A sum of cosines.**

1. Let $n \geq 2$. Deduce from the course that $\displaystyle\sum_{k=0}^{n-1} \cos\frac{2k\pi}{n} = 0$.
2. Write this equality for $n = 5$ and prove that $1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$.